LESSON PLAN

ENGINEERING METROLOGY (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Main rotor movements, dy-symmetry of lift feathering, flapping, dragging, hook's joint effect, flapping to equality, phase lag and advance angle concepts
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Main rotor movements, dy-symmetry of lift feathering, flapping, dragging, hook’s joint effect, flapping to equality, phase lag and advance angle concepts
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the helicopter control, main rotor movements, translation flight, power, auto rotation, vibration checks and various adjustments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Rotor Dynamics Diagrams; Safety: Do not touch rotating models; Instructor supervision required for diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary movements of the main rotor?
(b) Define the term 'flapping' in rotor dynamics.
(c) Why is phase lag critical in helicopter flight?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Main rotor movements types • Flapping and dragging mechanics • Hook's joint effect explanation • Flapping to equality principle • Phase lag and advance angle	L & Ds	PPT, IB	15 min
(b)	• Dynamic symmetry of lift • Feathering control inputs • Translation flight dynamics • Power management basics • Auto rotation entry	L & Ds	PPT, PDF	20 min
(c)	• Vibration check procedures • Adjustment mechanisms • Ground effect circulation • Recirculation effects analysis • Vertical descent techniques	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how phase lag affects rotor blade position relative to the hub.
(b) Describe the relationship between flapping and dragging in forward flight.
(c) How does dynamic symmetry of lift prevent excessive vibration?

12. Home Assignment and Follow Up	Prepare a brief note on the transition from ground effect to free air hover, referencing concepts from the next lesson plan (LP 06).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Hovering and forward flight, take off and climb to free air hover, vertical decent ground effect ground cushion re circulation effects, forces in balance during hover, forward flight and transition phenomenon of flap back, factors effecting maximum speed.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min